Ravensbourne University London

Project Title	Advanced Software Development
Deadline	August 7, 2026
Submission Method	Canvas
Project Staff	Dr Sermet Mir

Unit Title	Full Stack Application Development	**Level / Term**	5
Unit Code	CMS22204	**Credit / Weighting**	40 credits
Unit Leader	Dr Sermet Mir	**Date of Issue**	May 18, 2026

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Project Brief

Project Description:

In this comprehensive full-stack development project, Students will design and develop an open-ended full-stack web application that solves a real-world problem of their choice. The application should include a frontend interface, backend API, database integration, and at least one advanced feature such as authentication, search/filtering, or dashboard/reporting. Students may choose any suitable domain, such as task management, booking systems, e-commerce, learning platforms, or dashboards). The goal is to create a robust and scalable solution for a clear user or business need.

Project Deliverables:

1. Project Proposal:
   - Define the problem statement and the targeted user base.
   - Clearly outline the objectives and goals of the application.
   - Provide a high-level architecture diagram illustrating the components and data flow.
2. Front-End Development (Using React):
   - Develop a responsive and intuitive user interface to interact with the web application.
   - Implement state management using React state or context API.
   - Incorporate user authentication for secure access.
3. Back-End Development (Using Node.js):
   - Set up a Node.js server to handle HTTP requests and serve the React front-end.
   - Implement RESTful APIs to facilitate communication between the front end and the backend.
   - Integrate a Django ORM for database interactions and data persistence.
4. Database and Data Management:
   - Choose an appropriate database solution (SQL or NoSQL) for storing user data and application information.
   - Implement database schemas and models using the chosen database technology.
5. Core Features (Advanced Requirement):
   - Authentication and role-based access (admin/user, etc.).
   - Search, filtering and sorting functionalities.
   - Dashboard or reporting/analytics features.
6. Documentation:
   - Provide comprehensive documentation for setting up and running the application.
   - Include details about API endpoints, data models, and any additional configuration required.

Optional:

7. Deployment:
   - Deploy the full-stack application to a cloud platform (e.g., AWS, Heroku, or Azure).
   - Ensure scalability and reliability in the deployment environment.
8. Testing:
   - Implement unit tests for critical components.
   - Perform integration testing to validate the interaction between different modules.
9. Containerization using Docker:
   - Package the application using Docker to support portability and consistent deployment.

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Project Evaluation:

The project will be evaluated based on functionality, code quality, user experience, deployment efficiency, and adherence to best practices in full-stack development. Each student will present their project, demonstrating the application's features and overall system integration, including advanced features such as authentication, role-based access, search and filtering, and dashboard or reporting functionality.

Your project will include the following deliverables:

- A Multimedia project presentation of your fully working software solution containing link to the pre-recorded video and GitHub repo containing code files (10 minutes).
- A report demonstrating the project's objectives, system requirements, design, and implementation (500 words).

Essential and Recommended Reading List

Reading List

- Carl Rippon (2021) ASP.NET Core 5 and React: Full-stack web development using .NET 5, React 17, and TypeScript 4
- Jonathan and Hands (2021) Mastering Mobile Application Development, Learning iOS and Android side by side, Daniel Andres
- Pelaez Lopez, (2020), Full-Stack Web Development with Jakarta EE and Vue.js: Your One-Stop Guide to Building Modern Full-Stack Applications with Jakarta EE and Vue.js

Additional Reading List

- Full-Stack React, TypeScript, and Node: (2018) Build cloud-ready web applications using React 17 with Hooks and GraphQL(2020), Chris Northwood, Apress
- The Full Stack Developer: Your Essential Guide to the Everyday Skills Expected of a Modern Full Stack Web Developer

Assessment and tutor feedback date	TBD

Unit Learning outcomes (including Assessment Criteria)	
LO 1	Analyse the requirements of a live project or a self-initiated project
LO 2	Evaluate, short list and use appropriate tools required to complete the project
LO 3	Develop a high-quality artefact which covers the elements of a 'full-stack' development definition.
LO 4	Apply skills in documentation including project proposals, budgeting, selection of appropriate tools and environments.

Assessment Methods and Tasks		
Assessment tasks	**Submission Method**	**Weighting %**
• A Presentation of your fully working Software Project	i. Face to Face Demonstration	75%
• A report demonstrating the project's objectives, system requirements, design, and implementation (500 words). • With **mandatory** supplementary document of GitHub Repository containing code files	i. A canvas submission of Aula as 1 x PDF	25%

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The Five Principles	
The Five Principles underpin the Mindsets and Skillsets Manifesto and are the foundation upon which all course curriculum frameworks and unit specifications are based. The relevant Principles as stated below have been mapped against the Learning Outcomes relevant to each course unit and at each level (see Programme Specifications for full description of the Five Principles):	
Cultivate	Where the individual thrives.
Collaborate	Where the individual evolves.
Integrate	Where education engages industry.
Advocate	Where purpose meets practice.
Originate	Creativity meets technology.

Submission Regulations
Students who miss the deadline or fail this project will be required to complete a retrieval project which demonstrates that the learning outcomes have been achieved. Students should be aware that retrieval submissions are capped at a bare pass grade, (D-) unless the extenuating circumstances panel uphold an evidenced application. The deadline for retrieval submissions will be determined at an Assessment Board.